



Emam Hasan

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EDUCATION AND TRAINING

01/01/2020 – 30/09/2024 Dhaka, Bangladesh

BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND ENGINEERING United International University

Address United City, Madani Avenue, Badda, 1212, Dhaka, Bangladesh | **Website** <https://www.uiu.ac.bd> |

Field of study Artificial Intelligence, Machine Learning, Digital Image Processing, Statistics, Calculus and Linear Algebra, Data Structures and Algorithms

Final grade 3.57 out of 4.00 | **Type of credits** Trimester-based, credit-hour system | **Number of credits** 137 |

Thesis A Bilingual Bangla–Chittagonian Parallel Dataset for Low-Resource Machine Translation and Transformer-Based Modeling

WORK EXPERIENCE

MACHINE LEARNING ENGINEER – NASCENIA – 08/07/2024 – Current – LALMATIA, DHAKA, BANGLADESH

Website: <https://nascenia.com>

- Developed RAG AI chatbot backend leveraging LangChain, ChromaDB, Redis, LLaMA, Bangla NLU, and embedding models.
- Fine-tuned BanglaT5 model on 2.5M Bengali-English pairs using Hugging Face and OpenNMT.
- Deployed LLM, embedding, and translation models on GPU server in a Dockerized Python backend with REST APIs for RAG and NLP workflows.
- Built a Python-Flask web-scraping tool to automatically update the chatbot knowledge base.
- Built high-accuracy semantic search in a chatbot using embeddings, a re-ranker, and LLM-based prompt engineering to generate query variations for improved course and topic retrieval.
- Integrated AI chatbots into platforms like Messenger.
- Developed automated RAG tests for chatbot validation and version performance tracking.
- Worked with cross-functional teams to develop an intuitive, user-friendly chatbot.
- **Industry Projects:** Chatbot for [MyGov](#), [Muktopaath](#), [Teachers' Portal](#), and [ISEC](#) platforms.

PUBLICATIONS

2026

[Deep Learning-Enhanced OCT Image Analysis Pipeline: Integrating Denoising, Super-Resolution, and Fuzzy Logic for Improved Clinical Diagnostics](#)

Journal Name: Data Science, AI and Applications (ICDSAIA 2025) | **Publisher:** Springer

2025

[CellRegNet: Deep Learning for Protein Function in Cell Cycles](#)

Volume, Issue and Pages: Vol. 9, No. 1, June 2025, pp. 151-163 | **Publisher:** Bangladesh Research Foundation Journal

2024

[ThreatScan-Strengthening Website Security with Predictive Analytics Using Machine Learning Model](#)

Journal Name: 2024 IEEE Region 10 Symposium (TENSYP) | **Publisher:** IEEE

2024

[A Novel Ensemble Deep Learning Approach for Robust Multi-class Hair Fall Disease Classification](#)

Journal Name: 2024 IEEE Region 10 Symposium (TENSYP) | **Publisher:** IEEE

2023

Data Segmentation with Improved K-Means Clustering Algorithm

Journal Name: International Conference on Computer and Information Technology (ICCIT) | **Publisher:** IEEE

2023

Toward Lightweight Diabetic Retinopathy Classification: A Knowledge Distillation Approach for Resource Constrained Settings

Publisher: MDPI

2023

EWasteNet: A Two-Stream Data Efficient Image Transformer Approach for E-Waste Classification

Journal Name: 2023 IEEE 8th International Conference On Software Engineering and Computer Systems (ICSECS) | **Publisher:** IEEE

PROJECTS

01/01/2026 – 31/01/2026

Vehicle Detector

Personal Project | *Python, YOLO, ONNX, Android Runtime*

- Trained a lightweight YOLOv26n model on a custom labeled dataset to detect 11 vehicle classes.
- Converted the model to ONNX and deployed it on Android using ONNX Runtime for on-device inference.
- Achieved 22 FPS (45 ms latency) optimized for edge performance.

Link <https://github.com/emamhasanai/vehicle-detection>

01/06/2024 – 01/09/2024

NewsTok

Team Project (2-person) | *Spring Boot, Python, Flask, Selenium, JWT, REST APIs*

- Built TikTok-style news platform converting scraped articles into AI-generated short reels.
- Implemented dual-server architecture (admin + user) with real-time news scraping.
- Secured with Spring Security and JWT; managed data using Spring Data JPA.

Link <https://github.com/PaleBlueDot-repo>

01/01/2024 – 29/02/2024

Cinephiles

Team Lead (5 developers) | *Laravel, PHP, JavaScript, MySQL*

- Developed full-stack movie platform with recommendation engine, watch parties, and social features.
- Led team of 5 developers, managed task distribution and code reviews.

Link https://github.com/emamhasanai/cinephiles_

SKILLS

Programming

Python, Java, C++, JavaScript, SQL

Natural Language Processing (NLP)

Hugging Face, Transformers, LangChain, Embeddings, RAG Pipelines, Knowledge Distillation

Machine Learning

PyTorch, NumPy, Pandas, Scikit-learn, Vector Databases, Classification, Clustering

Computer Vision & Image Processing

Deep Learning for Image Analysis and Classification, Inference Optimization (Latency/Cost)

Tools/Frameworks

AWS, Git, Trello, RESTful APIs, Docker, Flask, Selenium, Playwright, BeautifulSoup

Databases

MySQL, ChromaDB, Redis

Soft Skills

Problem-Solving, Team Collaboration, Communication

● LANGUAGE SKILLS

Mother tongue(s): **BENGALI**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	B2	B2	B2	B2	C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● HONOURS AND AWARDS

08/03/2024

National Collegiate Programming Contest – Jahangirnagar University

- Finalist

01/10/2023

Research Grant – United International University

- Received 2500 USD research grant from Institute of Advanced Research

09/01/2023

UIU CSE project Show – United International University

- 2nd Runner-up in Microprocessors and Micro-controllers Lab Category of CSE project Show
- 3rd Runner-up in System Analysis and Design Lab Category of CSE project Show

01/12/2022

UIU Intra University AI Contest – United International University

- 1st place out of 100+ participants

01/08/2022

UIU Coders’ Combat 3.0 – United International University

- Secured 4th place out of 200 participants

Link <https://www.newagebd.net/article/177489/uiu-coders-combat-30-concludes>